

TOWARDS COHERENT VIDEO COLORIZATION: WHEN OPTICAL FLOW MEETS IMAGE DIFFUSION MODELS - SUPPLEMENTARY MATERIAL

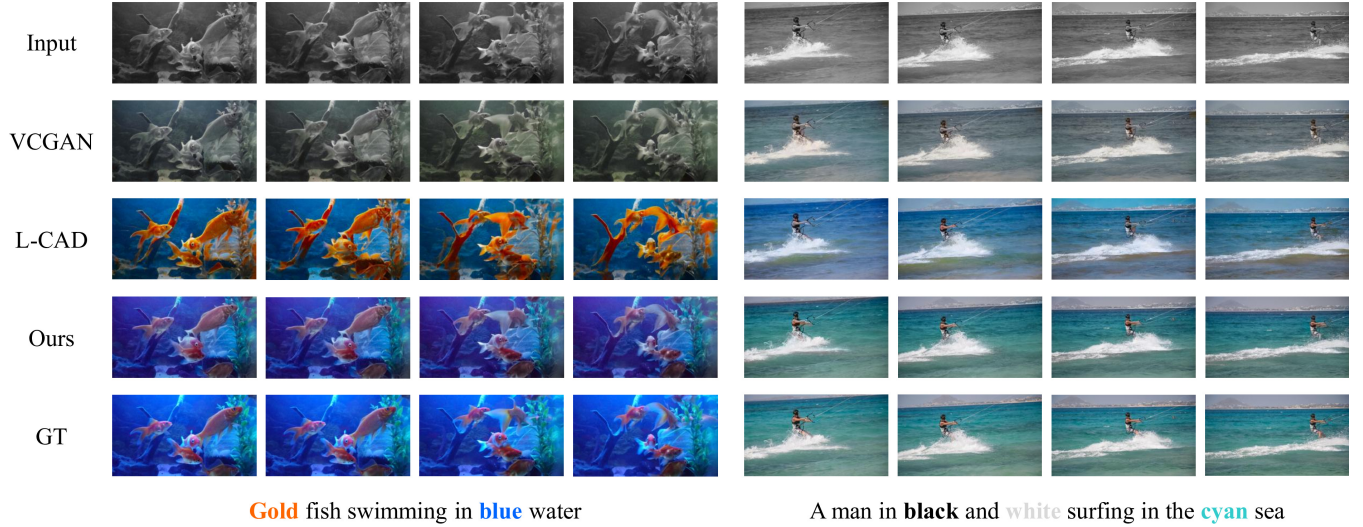


Fig. 1. Additional qualitative comparison results. Our model performs better semantic reconstruction and color assignment.

1. ADDITIONAL EXPERIMENTS

1.1. Qualitative Comparison

As shown in Fig. 1, our method achieves the best inter-frame color consistency, along with realistic and faithful reconstruction of the original video than VCGAN [1] and L-CAD [2]. On the left, our model captures the original color distribution of the goldfish under the disturbance of the blue water environment, correctly assigning the colors and maintaining better semantic accuracy. On the right, our method recolors the sea in better alignment with the ground truth; for the sky, which is unmentioned in the colorization prompt, our model correctly extracts the grayish color from the reference frame instead of coloring it blue as L-CAD [2] does, meanwhile remaining consistent across frames.

1.2. Ablation on Hybrid Training Scheme

We further verify the effectiveness of our hybrid training scheme via ablation studies. We conduct experiments using the checkpoint after optical flow pretraining phase only and both phases, and test on an additional set of generated video samples. Results are shown in Table 1.

Optical flow pretraining significantly boosts metrics over the baseline, confirming effective motion pattern learning. The performance on real video data remains mostly consistent with that on generated data, and clearly surpasses the baseline and the first stage model, indicating no overfitting to generated videos and demonstrating effective mitigation of the domain gap.

Table 1. Quantitative results of ablation studies on hybrid training scheme. Best and second best performance are highlighted in **bold** and *italic*, respectively.

Method	CDC↓	PSNR↑	SSIM↑	LPIPS↓
Baseline	3.902	20.18	0.873	0.226
Ours (first stage only)	3.651	25.78	0.935	0.222
Ours (on DAVIS)	<i>3.303</i>	<i>26.35</i>	<i>0.950</i>	<i>0.207</i>
Ours (on generated test set)	3.115	28.17	0.962	0.202

1.3. Runtime Analysis

We perform analysis of runtime, comparing with methods BiSTNet [3], L-CAD [2] and I2VEdit [4], averaged across the DAVIS dataset with a total of 10,474 frames:

Table 2. Runtime analysis on DAVIS dataset.

Method	BiSTNet	L-CAD	I2VEdit	Ours
Runtime(s/frame)	0.68	10.45	30.62	<i>3.56</i>

While slower than CNN-based BiSTNet, we are much faster than diffusion-based methods, successfully optimizing complexity and lowering computational cost with single-step inference design.

2. ANALYSIS OF FAILURE CASES

Two failure cases of our method are shown in Fig. 2. In the upper case, the motion in the video moves rapidly, causing

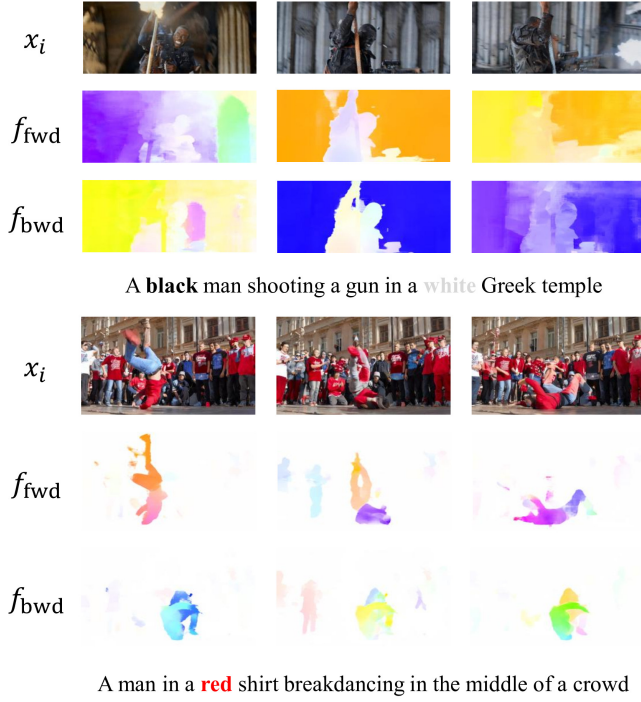


Fig. 2. Failure cases.

optical flow to be blurry and most of the scene to fall in occlusion, especially when the colorized frame is far away from the reference frame; this results in optical flow guidance and OFCFA being ineffective. In the below case, the prompt fails to accurately describe the whole complex scene with details, causing the model to wrongly assign colors. These failure cases are also common issues for other video colorization methods, revealing the core challenges in the field of video editing when dealing with large-scale motion and complex scene video inputs. Future work can focus on incorporating semantic guidance to further enhance integrity of complex structures and motions.

3. REFERENCES

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